

EXPLORING FINE-TUNING FOR TABULAR FOUNDATION MODELS

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Abstract

Tabular Foundation Models (TFMs) have recently shown strong in-context learning capabilities on structured data, achieving zero-shot performance comparable to traditional machine learning methods. We find that zero-shot TFMs already achieve strong performance, while the benefits of fine-tuning are highly model- and data-dependent. Meta-learning and PEFT provide moderate gains under specific conditions, whereas full supervised fine-tuning (SFT) often reduces accuracy or calibration quality. This work presents the first comprehensive study of fine-tuning in TFMs across benchmarks including TALENT, OpenML-CC18, and TabZilla. We compare Zero-Shot, Meta-Learning, Supervised (SFT), and parameter-efficient (PEFT) approaches, analyzing how dataset factors—such as imbalance, size, and dimensionality—affect outcomes. Our findings cover performance, calibration, and fairness, offering practical guidelines on when fine-tuning is most beneficial and its limitations.

Keywords: Tabular Foundation Models, Meta-Learning, PEFT, Calibration, Fairness, Benchmarking.



1 Introduction

Tabular data drives applications in enterprises like healthcare, finance, and analytics. For decades, gradient-boosted decision trees (GBDTs) such as XGBoost [1], LightGBM [2], and CatBoost [3] have dominated due to their robustness on heterogeneous features and small datasets. While deep learning has revolutionized vision and NLP, standard neural networks often struggle to outperform GBDTs on tabular tasks [4, 5], though recent architectures like FT-Transformer [6] and TabR [7] have narrowed this gap.

Tabular Foundation Models (TFMs) offer a new paradigm, leveraging large-scale pretraining to enable in-context learning (ICL) and zero-shot generalization. TABPFN [8, 9] approximates Bayesian inference using transformers trained on synthetic data, effectively performing posterior inference in a single forward pass for small datasets. Building on this, TABICL [10] scales ICL to larger datasets via a two-stage column-then-row attention mechanism. To address high-dimensional dependencies, ORIONMSP [11] and ORIONBIX [12] introduce multi-scale sparse attention and biaxial attention mechanisms, respectively, enabling efficient context modeling. TABDPT [13] employs diffusion-based pretraining to learn robust representations, while MITRA [14] utilizes mixed synthetic priors to improve generalization.

While zero-shot inference is powerful, adapting TFMs to target distributions can enhance performance. META-LEARNING (episodic fine-tuning) preserves ICL capabilities by training on support-query splits, mimicking the inference-time environment [15]. SUPERVISED FINE-TUNING (SFT) updates all parameters on labeled data [16], which can yield gains on large datasets but risks overfitting on small ones. PARAMETER-EFFICIENT FINE-TUNING (PEFT), particularly Low-Rank Adaptation (LoRA) [17], offers a resource-efficient alternative by updating only low-dimensional

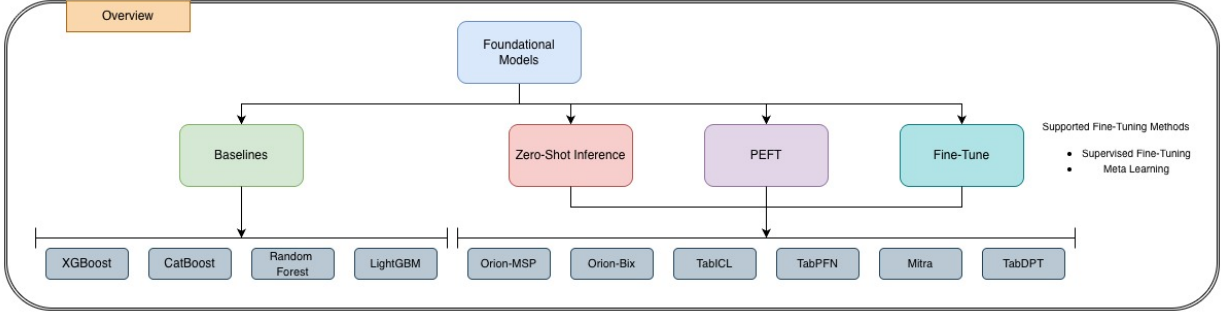


Figure 1 High-level overview of models and adaptation strategies. We benchmark classical tabular baselines and multiple tabular foundation models under three regimes—zero-shot inference, parameter-efficient fine-tuning (PEFT), and full supervised fine-tuning—using both supervised fine-tuning and meta-learning variants

subspaces, often matching full fine-tuning performance with lower overhead [18, 19]. Strategies like LOCALPFN (Thomas et al. [20]) have also emerged, combining retrieval with fine-tuning to adapt to local data sub-manifolds.

Beyond predictive performance, responsible deployment requires assessing model trustworthiness through calibration and fairness. Calibration ensures that predicted probabilities align with true outcomes, a critical requirement for high-stakes decision-making in healthcare and finance. Fairness necessitates that models avoid systematic bias across demographic groups. While TFMs show promise in zero-shot settings, the impact of different adaptation strategies on these safety-critical dimensions remains under-explored. However, it remains unclear *when* fine-tuning helps these models, *which* strategies are effective, and *how* dataset properties influence outcomes. The wide variation in tabular dataset size, imbalance, and feature structure makes the impact of fine-tuning particularly non-obvious. This work presents the first large-scale study of fine-tuning TFMs.

Our contributions are: (I) a unified evaluation of four adaptation strategies—zero-shot, meta-learning, supervised fine-tuning (SFT), and parameter-efficient fine-tuning (PEFT)—across six TFMs and multiple benchmarks; (II) an analysis of how dataset factors (size, imbalance, dimensionality) shape fine-tuning outcomes; (III) an assessment of calibration and fairness effects; and (IV) practical guidelines on when fine-tuning is effective. Our findings show that fine-tuning is not universally beneficial, underscoring the need for careful strategy selection. An Overview of all the models and adaptation strategies is shown in Figure 1

2 Methodology

2.1 Models and Benchmarks

We evaluate six **Tabular Foundation Models**—TabPFN, TabICL, OrionMSP, OrionBiX, TabDPT, and Mitra—alongside classical baselines (XGBoost, LightGBM, CatBoost, Random Forest). Experiments span three benchmark suites: TALENT (155 datasets) [21], OPENML-CC18 (63 datasets) [22], and TABZILLA (27 datasets) [23]. For fairness evaluation, we additionally use nine public datasets with sensitive attributes, including *Adult Census Income*, *German Credit*, and *COMPAS Recidivism*. All experiments follow standardized splits and are executed on NVIDIA L40S and H200 GPUs. We utilize TabTune [24] for TFM experiments and its AutoGluon [25] implementation for baselines.

2.2 Adaptation Strategies

We compare four distinct adaptation paradigms to assess their impact on performance, calibration, and fairness:

- **Zero-Shot Inference:** Evaluates pretrained TFMs directly on target tasks without parameter updates, relying solely on in-context learning or prior knowledge.
- **Meta-Learning Fine-Tuning:** Employ episodic training (48–512 support samples per episode for 5 epochs) to adapt the model while preserving its in-context learning capabilities.
- **Supervised Fine-Tuning (SFT):** Performs full-parameter optimization using the AdamW optimizer with learning rates between $1e^{-5}$ and $5e^{-5}$, employing early stopping (up to 10 epochs) to mitigate overfitting.
- **Parameter-Efficient Fine-Tuning (PEFT):** Utilizes Low-Rank Adaptation (LoRA) [17] with rank $r=8$ and $\alpha=16$. This strategy is applied to both supervised and meta-learning regimes to reduce memory overhead while maintaining adaptation flexibility.

Table 1 Overall leaderboard across three benchmark suites—TALENT, OpenML-CC18, and TabZilla. Ranks denote the mean rank per benchmark suite (lower is better). Metrics: ACC = Accuracy, F1 = Weighted F1. The "All" column reports the aggregated rank across for a strategy. Models are grouped by adaptation strategy. Formatting: **1st place**; 2nd place within each group. *Note : that rank reflects only relative ordering and may not capture absolute ACC/F1 trends, which can vary across scenarios.*

Strategy	All	TALENT			OpenML-CC18			TabZilla		
Models	Rank(↓)	Rank(↓)	ACC(↑)	F1(↑)	Rank(↓)	ACC(↑)	F1(↑)	Rank(↓)	ACC(↑)	F1(↑)
<i>Baselines + Zero-Shot Inference</i>										
XGBoost	6.39	5.97	0.8403	0.8360	5.34	0.8558	0.8537	5.62	0.8612	0.8326
CatBoost	6.16	5.54	0.8336	0.8259	6.64	0.8588	0.8520	5.40	0.8579	0.8384
Random Forest	7.06	6.11	0.8285	0.8209	5.78	0.8547	0.8497	7.72	0.8358	0.8399
LightGBM	6.50	6.07	0.8331	0.8245	5.60	0.8581	0.8493	4.90	0.8618	0.8211
TabICL	4.78	4.07	<u>0.8471</u>	<u>0.8379</u>	4.36	0.8667	0.8623	5.63	0.8734	0.8698
OrionBiX	5.27	4.56	0.8346	0.8260	4.97	0.8653	0.8596	4.63	0.8728	0.8628
OrionMSP	3.76	3.26	0.8461	0.8360	3.90	0.8722	0.8676	3.78	0.8821	0.8786
TabPFN	4.44	<u>3.71</u>	0.8514	0.8412	4.36	<u>0.8714</u>	<u>0.8663</u>	4.65	0.8752	0.8716
Mitra	10.85	9.67	0.3921	0.2868	9.55	0.3614	0.2522	10.21	0.3152	0.1830
TabDPT	5.27	5.16	0.8408	0.8318	<u>4.32</u>	0.8672	0.8625	<u>3.81</u>	<u>0.8814</u>	<u>0.8775</u>
<i>FineTune – Meta Learning</i>										
TabICL	3.66	3.33	0.8344	0.8253	3.05	0.8664	0.8597	4.15	0.6956	0.6845
OrionBiX	3.96	4.25	0.8158	0.8060	2.86	<u>0.8548</u>	<u>0.8516</u>	2.00	<u>0.8726</u>	<u>0.8662</u>
OrionMSP	2.26	1.80	0.8401	0.8310	2.82	<u>0.8548</u>	<u>0.8516</u>	1.73	0.8735	0.8672
TabPFN	<u>2.42</u>	<u>2.07</u>	0.8517	0.8414	2.42	0.8842	0.8784	2.42	0.8663	0.8603
Mitra	6.09	5.79	0.6416	0.5874	5.42	0.6164	0.5651	4.70	0.5592	0.5147
TabDPT	3.95	3.72	0.8255	0.8167	4.40	0.8534	0.8501	2.62	0.8500	0.8456
<i>FineTune – Supervised Fine Tuning</i>										
TabICL	5.05	4.74	0.7668	0.7439	4.36	0.6838	0.6299	4.65	0.5670	0.4733
OrionBiX	4.21	3.54	0.7698	0.7469	4.30	0.7126	0.6595	4.25	0.6476	0.5782
OrionMSP	2.88	2.27	0.7908	0.7653	3.18	0.7995	0.7668	2.88	0.7454	0.7222
TabPFN	1.97	1.83	0.8459	0.8350	1.89	0.8697	0.8617	1.86	0.8433	0.8327
Mitra	5.98	5.62	0.5460	0.4382	5.02	0.5408	0.4309	5.52	0.4608	0.3467
TabDPT	<u>2.79</u>	2.98	0.8202	0.8094	<u>2.25</u>	<u>0.8499</u>	<u>0.8424</u>	1.81	<u>0.8337</u>	<u>0.8260</u>
<i>FineTune – PEFT - Meta Learning</i>										
TabICL	4.32	4.07	0.7017	0.6867	4.07	0.7773	0.7605	3.45	0.7116	0.7003
OrionBiX	2.77	2.39	0.7854	<u>0.7762</u>	2.50	0.8471	0.8430	3.52	0.7370	0.7200
OrionMSP	2.21	<u>2.25</u>	<u>0.7879</u>	0.7728	1.93	<u>0.8566</u>	0.8453	1.33	0.8594	0.8581
Mitra	4.64	4.17	0.6369	0.5905	4.32	0.6000	0.5426	4.36	0.5461	0.4960
TabDPT	<u>2.28</u>	2.11	0.8002	0.7910	<u>2.17</u>	0.8600	0.8539	2.33	<u>0.8495</u>	<u>0.8481</u>
<i>FineTune – PEFT - Supervised Fine Tuning</i>										
TabICL	3.70	3.72	0.5692	0.4638	3.06	0.8174	0.7965	2.92	0.7920	0.7776
OrionBiX	3.99	3.57	0.6358	0.5570	3.67	0.7380	0.6789	3.95	0.6707	0.6071
OrionMSP	2.01	2.01	0.6749	0.5956	1.73	<u>0.8241</u>	<u>0.8033</u>	1.47	<u>0.8214</u>	<u>0.8071</u>
Mitra	4.62	4.01	0.5294	0.4303	4.49	0.4965	0.3917	4.70	0.4606	0.3597
TabDPT	1.91	1.65	0.7988	0.7842	2.03	0.8500	0.8398	<u>1.95</u>	0.8461	0.8461

2.3 Evaluation Metrics

Our assessment covers three complementary dimensions:

- **Performance:** Measured via Accuracy, weighted F1-score, and mean rank across datasets.
- **Calibration:** Assessed using Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and Brier Score [26] to quantify the reliability of probability estimates.
- **Fairness:** Evaluated using Statistical Parity Difference (SPD), Equalized Odds Difference (EOD), and Equal Opportunity Difference (EOpD) [27, 28, 29] to detect group disparities.

2.4 Research Questions

In this work, we study: (1) **whether fine-tuning improves TFM performance compared to zero-shot inference**; (2) **how dataset properties such as size, imbalance, and dimensionality influence these gains**; (3) **how different adaptation strategies affect model calibration**; and (4) **the implications of fine-tuning on fairness**.

Table 2 Comprehensive performance analysis across dataset characteristics: size (Small <1K, **Medium 1K–10K**, Large >10K samples), class balance (Balanced ≥ 0.6 , Imbalanced <0.6), and feature dimensionality (Narrow <10, **Medium 10–100**, **Wide >100 features**). ACC = Accuracy; F1 = Weighted F1-score. Values are averaged across datasets within each category. Models are grouped by adaptation strategy. Formatting: **1st place**; 2nd place within each group.

Strategy	Dataset Size						Dataset Imbalance				Dataset Width					
	Small (<1K)		Medium (1K-10K)		Large (>10K)		Balanced (≥ 0.6)		Imbalanced (<0.6)		Narrow (<10)		Medium (10-100)		Wide (>100)	
Models	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)	ACC($\uparrow$)	F1($\uparrow$)
<i>Baselines + Zero-Shot Inference</i>																
XGBoost	0.8168	0.7964	0.8363	0.8314	0.8969	0.8920	0.8175	0.8110	0.8859	0.8785	0.8222	0.8159	0.8482	0.8410	0.9140	0.9039
CatBoost	0.8124	0.7935	0.8340	0.8264	0.8797	0.8733	0.8076	0.8020	0.8785	0.8665	0.8145	0.8067	0.8441	0.8344	0.9157	0.9084
Random Forest	0.7988	0.8187	0.8285	0.8221	0.8694	0.8628	0.7983	0.7955	0.8741	0.8646	0.8005	0.7044	0.8410	0.8235	0.9034	0.8936
LightGBM	0.8143	0.7789	0.8314	0.8226	0.8827	0.8764	0.8071	0.7977	0.8775	0.8633	0.8128	0.7907	0.8458	0.8326	0.8999	0.8908
TabICL	0.8301	0.8338	0.8486	0.8398	0.8802	0.8743	0.8279	0.8233	0.8806	0.8698	0.8208	0.8119	0.8627	0.8549	0.9101	0.8936
OrionBiX	0.8330	0.8150	0.8348	0.8260	0.8729	0.8670	0.8096	0.8040	0.8787	0.8683	0.8112	0.8043	0.8510	0.8417	0.8859	0.8849
OrionMSP	0.8232	0.8194	0.8494	0.8402	0.8843	0.8768	0.8265	0.8202	0.8840	0.8731	0.8394	0.8314	0.8572	0.8478	0.8860	0.8837
TabPFN	0.8325	0.8131	0.8557	0.8462	0.8783	0.8713	0.8367	0.8309	0.8808	0.8697	0.8187	0.8092	0.8676	0.8589	0.9129	0.9111
Mitra	0.4334	0.3236	0.3600	0.2553	0.3837	0.2754	0.2763	0.1540	0.4794	0.3858	0.3737	0.2683	0.3886	0.2781	0.2521	0.1497
TabDPT	0.8333	0.8271	0.8424	0.8339	0.8831	0.8765	0.8233	0.8189	0.8798	0.8690	0.8262	0.8189	0.8566	0.8483	0.8845	0.8820
<i>FineTune – Meta Learning</i>																
TabICL	0.7912	0.7806	0.8342	0.8256	0.8431	0.8360	0.7943	0.7884	0.8662	0.8547	0.7867	0.7779	0.8383	0.8338	0.8319	0.8667
OrionBiX	0.8149	0.8123	0.8283	0.8204	0.8410	0.8306	0.7947	0.7881	0.8622	0.8532	0.7956	0.7900	0.8316	0.8278	0.8429	0.8640
OrionMSP	0.8211	0.8151	0.8413	0.8330	0.8787	0.8721	0.8231	0.8183	0.8735	0.8636	0.8253	0.8190	0.8514	0.8457	0.8564	0.8737
TabPFN	0.8336	0.8256	0.8638	0.8548	0.8748	0.8668	0.8461	0.8407	0.8784	0.8664	0.8264	0.8171	0.8683	0.8579	0.9172	0.9533
Mitra	0.6417	0.5988	0.6083	0.5516	0.6585	0.6132	0.4699	0.4142	0.7885	0.7391	0.6646	0.6247	0.6004	0.5763	0.4090	0.3224
TabDPT	0.8224	0.8188	0.8289	0.8212	0.8682	0.8613	0.8066	0.8023	0.8650	0.8552	0.8129	0.8071	0.8405	0.8350	0.8400	0.8497
<i>FineTune - Supervised FineTuning</i>																
TabICL	0.6746	0.6225	0.7553	0.7187	0.6693	0.6212	0.6708	0.6317	0.7794	0.7362	0.6609	0.6195	0.7507	0.7101	0.7757	0.7269
OrionBiX	0.7158	0.6792	0.7718	0.7386	0.6697	0.6272	0.6917	0.6550	0.7934	0.7590	0.6922	0.6576	0.7643	0.7294	0.7560	0.7046
OrionMSP	0.8135	0.7920	0.8036	0.7798	0.7231	0.6828	0.7467	0.7287	0.8307	0.7953	0.7540	0.7297	0.8025	0.7750	0.8148	0.7808
TabPFN	0.8216	0.8109	0.8580	0.8485	0.8572	0.8459	0.8336	0.8267	0.8709	0.8578	0.8258	0.8162	0.8578	0.8469	0.9346	0.9335
Mitra	0.5625	0.4408	0.5133	0.4116	0.5720	0.4501	0.3216	0.2037	0.7413	0.6415	0.5619	0.4478	0.5349	0.4268	0.3307	0.2436
TabDPT	0.8227	0.8153	0.8377	0.8284	0.8120	0.8002	0.8025	0.7949	0.8578	0.8464	0.8078	0.7948	0.8378	0.8294	0.8623	0.8587
<i>FineTune – PEFT - Meta Learning</i>																
TabICL	0.7518	0.7397	0.7251	0.7114	0.6962	0.6799	0.6530	0.6460	0.8026	0.7800	0.6375	0.6375	0.7601	0.7444	0.7714	0.7622
OrionBiX	0.8093	0.8042	0.7995	0.7939	0.7848	0.7665	0.7504	0.7436	0.8528	0.8420	0.7584	0.7584	0.8147	0.8057	0.8251	0.8243
OrionMSP	0.8275	0.8213	0.8134	0.8019	0.8013	0.7817	0.7714	0.7623	0.8608	0.8433	0.7836	0.7836	0.8254	0.8107	0.8293	0.8256
Mitra	0.6510	0.6012	0.6089	0.5599	0.6304	0.5795	0.4683	0.4173	0.8001	0.7524	0.6471	0.6471	0.6177	0.5650	0.4129	0.3338
TabDPT	0.8191	0.8157	0.8121	0.8049	0.8407	0.8299	0.7860	0.7801	0.8609	0.8511	0.7979	0.7979	0.8299	0.8209	0.8416	0.8376
<i>FineTune - PEFT - Supervised FineTuning</i>																
TabICL	0.7957	0.7710	0.6424	0.5686	0.6084	0.5020	0.6003	0.5233	0.7223	0.6507	0.6196	0.5287	0.6596	0.5902	0.8760	0.8502
OrionBiX	0.7576	0.7102	0.6585	0.5892	0.6327	0.5381	0.5974	0.5164	0.7490	0.6875	0.6278	0.5444	0.6851	0.6177	0.6737	0.6166
OrionMSP	0.8298	0.8132	0.7190	0.6597	0.6931	0.6152	0.6753	0.6105	0.7915	0.7423	0.6795	0.6113	0.7477	0.6939	0.8042	0.7663
Mitra	0.5244	0.4133	0.5008	0.4013	0.5459	0.4476	0.3189	0.2171	0.7495	0.6497	0.5518	0.4481	0.5106	0.4108	0.2831	0.1874
TabDPT	0.8269	0.8182	0.8332	0.8245	0.7686	0.7418	0.7907	0.7798	0.8496	0.8346	0.7821	0.7631	0.8305	0.8201	0.8925	0.8901

Table 3 Probability calibration metrics across models and tuning strategies on the TALENT, OpenML-CC18, and TabZilla benchmark suites. Values are averaged across datasets within each suite. Models are grouped by adaptation strategy. Formatting: **1st place** ; **2nd place** within each group.

Strategy	TALENT			OpenML-CC18			TabZilla		
Models	ECE(↓)	MCE(↓)	BS(↓)	ECE(↓)	MCE(↓)	BS(↓)	ECE(↓)	MCE(↓)	BS(↓)
<i>Zero-Shot Inference</i>									
TabICL	0.0219	0.2421	0.1533	0.0371	0.3404	0.1267	0.0369	0.2863	0.1301
OrionBiX	0.0324	0.2245	0.1787	0.0325	0.3230	0.1325	0.0385	0.2512	0.2419
OrionMSP	0.0219	0.2098	0.1589	0.0319	0.2902	0.1262	0.0310	0.2805	0.1243
TabPFN	0.0276	0.2470	0.1514	0.0375	0.2880	0.1253	0.0431	0.2745	0.1283
Mitra	0.2238	0.3115	0.5291	0.2138	0.2952	0.5307	0.1946	0.2995	0.5733
TabDPT	0.0308	0.2489	0.1586	0.0443	0.3160	0.1351	0.0435	0.2997	0.1227
<i>FineTune - Meta Learning</i>									
TabICL	0.0355	0.2570	0.1671	0.0426	0.3387	0.1346	0.2033	0.4926	0.3732
OrionBiX	0.0409	0.2693	0.1841	0.0731	0.4164	0.1607	0.0744	0.3964	0.1416
OrionMSP	0.0340	0.2570	0.1656	0.0520	0.3458	0.1606	0.0365	0.2574	0.1250
TabPFN	0.0271	0.2440	0.1502	0.0346	0.2989	0.1178	0.0388	0.2636	0.1344
Mitra	0.0334	0.1912	0.3479	0.0533	0.2258	0.3932	0.0891	0.2086	0.4239
TabDPT	0.0562	0.3235	0.1791	0.0745	0.4178	0.1620	0.0664	0.2838	0.1591
<i>FineTune - Supervised FineTuning</i>									
TabICL	0.0768	0.3219	0.2349	0.1156	0.3610	0.3039	0.1534	0.3280	0.4078
OrionBiX	0.0753	0.3204	0.2334	0.1131	0.3374	0.2923	0.1581	0.3801	0.3530
OrionMSP	0.0471	0.1886	0.2113	0.0550	0.3155	0.1995	0.0773	0.3321	0.2594
TabPFN	0.0287	0.2509	0.1580	0.0381	0.2957	0.1246	0.0469	0.2528	0.1454
Mitra	0.1723	0.2620	0.4677	0.1539	0.2269	0.4784	0.2023	0.2991	0.5254
TabDPT	0.0548	0.3019	0.1867	0.0497	0.3397	0.1443	0.0713	0.3183	0.1801
<i>PEFT FineTune - Meta Learning</i>									
TabICL	0.0355	0.2570	0.1671	0.0426	0.3387	0.1346	0.2033	0.4926	0.3732
OrionBiX	0.0409	0.2693	0.1841	0.0731	0.4164	0.1607	0.0744	0.3964	0.1416
OrionMSP	0.0340	0.2570	0.1656	0.0520	0.3458	0.1606	0.0365	0.2574	0.1250
Mitra	0.0334	0.1912	0.3479	0.0533	0.2258	0.3932	0.0891	0.2086	0.4239
TabDPT	0.0562	0.3235	0.1791	0.0745	0.4178	0.1620	0.0664	0.2838	0.1591
<i>PEFT FineTune - Supervised FineTuning</i>									
TabICL	0.0768	0.3219	0.2349	0.1156	0.3610	0.3039	0.1534	0.3280	0.4078
OrionBiX	0.0753	0.3204	0.2334	0.1131	0.3374	0.2923	0.1581	0.3801	0.3530
OrionMSP	0.0471	0.1886	0.2113	0.0550	0.3155	0.1995	0.0773	0.3321	0.2594
Mitra	0.1723	0.2620	0.4677	0.1539	0.2269	0.4784	0.2023	0.2991	0.5254
TabDPT	0.0548	0.3019	0.1867	0.0497	0.3397	0.1443	0.0713	0.3183	0.1801

3 Experimental Results

We assess zero-shot and fine-tuned TFMs across performance, calibration, and fairness. Table 1 summarizes results on TALENT, OpenML-CC18, and TabZilla.

Across accuracy and F1, zero-shot inference remains competitive and often outperforms fine-tuned variants, with several models showing reduced performance after SFT. Across all suites, zero-shot TFMs already achieve strong accuracy and F1, often surpassing or matching their fine-tuned variants. OrionMSP and TabPFN provide the most competitive zero-shot results. Meta-learning offers moderate and architecture-dependent gains, particularly for TabPFN and OrionMSP, but full supervised fine-tuning (SFT) frequently reduces accuracy and F1—most notably for TabICL and OrionBiX. PEFT variants recover part of this lost performance and typically track meta-learning behaviour while retaining their computational efficiency, with TabDPT showing the most stable improvements under PEFT. Models responses to adaptation vary, reflecting the distinct pretraining processes behind each architecture.

3.1 Performance Across Dataset Characteristics.

Table 2 breaks down performance by dataset size, imbalance, and feature dimensionality. Fine-tuning is most beneficial on medium-sized datasets (1K–10K samples), where TabPFN and OrionMSP show small but consistent gains. On small datasets (<1K), zero-shot inference remains clearly superior, as most models overfit when fine-tuned. On large datasets (>10K), adaptation provides limited benefit beyond zero-shot predictions, with OrionMSP and TabDPT already performing strongly. For imbalanced datasets, meta-learning offers the most reliable improvements; SFT continues to degrade several architectures. Wide-feature datasets (>100 features) exhibit the few cases where fine-tuning

Table 4 Fairness metrics across models and tuning strategies on datasets with explicit demographic attributes. Values are averaged across fairness evaluation datasets. Models are grouped by adaptation strategy. Formatting: 1st place ; 2nd place within each group.

Strategy	Fairness Metrics			Performance Metrics	
Models	SPD($\downarrow$)	EOD($\downarrow$)	EOPD($\downarrow$)	ACC($\uparrow$)	F1($\uparrow$)
<i>Zero-Shot Inference</i>					
TabICL	0.2900	0.3168	0.3278	0.8743	0.8680
OrionBiX	0.3328	0.2698	0.2855	0.8779	0.8727
OrionMSP	0.3380	0.2827	0.2983	<u>0.8752</u>	<u>0.8706</u>
TabPFN	0.3070	0.3114	0.3176	0.8708	0.8631
Mitra	0.0193	0.0590	0.0982	0.6902	0.5847
TabDPT	0.2990	0.3197	0.3338	0.8674	0.8582
<i>FineTune - Meta Learning</i>					
TabICL	0.3140	0.3100	0.3332	0.8678	0.8603
OrionBiX	0.2021	0.1624	0.2010	0.8743	0.8692
OrionMSP	0.3038	0.2798	<u>0.2907</u>	0.8680	0.8650
TabPFN	0.3070	0.3115	0.3177	<u>0.8733</u>	<u>0.8668</u>
Mitra	0.2586	0.2748	0.3301	0.7967	0.7546
TabDPT	0.2648	0.3180	0.3080	0.8614	0.8503
<i>FineTune - Supervised FineTuning</i>					
TabICL	<u>0.0550</u>	0.3225	0.3319	0.4584	0.3625
OrionBiX	0.1128	<u>0.1761</u>	<u>0.1817</u>	0.6180	0.5498
OrionMSP	0.1227	0.2208	0.2353	0.4753	0.4039
TabPFN	0.3070	0.3115	0.3177	0.8733	0.8668
Mitra	0.0161	0.0170	0.0555	0.7268	0.6277
TabDPT	0.3075	0.3153	0.3284	<u>0.8529</u>	<u>0.8435</u>
<i>PEFT FineTune - Meta Learning</i>					
TabICL	0.3140	0.3092	0.3332	0.8678	0.8602
OrionBiX	0.3149	0.2722	0.2875	0.8748	0.8717
OrionMSP	0.3038	0.2798	<u>0.2907</u>	0.867	<u>0.8632</u>
Mitra	0.2595	0.2391	0.3154	0.8104	0.7721
TabDPT	<u>0.2778</u>	0.2792	0.3051	0.8651	0.8556
<i>PEFT FineTune - Supervised FineTuning</i>					
TabICL	0.1572	0.4041	0.4089	0.4881	0.4269
OrionBiX	<u>0.0349</u>	<u>0.1564</u>	<u>0.2234</u>	0.6401	0.5759
OrionMSP	0.1132	0.2140	0.2420	0.5732	0.5080
Mitra	0.0029	0.0370	0.0740	0.7133	0.6028
TabDPT	0.2960	0.3190	0.3342	0.8595	0.8505

provides noticeable boosts, such as for TabPFN (SFT). These mixed effects within each regime indicate that dataset heterogeneity—feature correlations, noise, and distributional skew—strongly shapes how models respond to adaptation.

3.2 Calibration Evaluation :

As shown in Table 3, zero-shot TFMs deliver the best overall calibration, with OrionMSP and TabPFN achieving the lowest ECE values across all benchmark suites. Meta-learning largely preserves this reliability while providing modest performance gains. In contrast, SFT substantially worsens calibration for most transformers, increasing both ECE and MCE; TabPFN is the only model that remains consistently well-calibrated after full fine-tuning. PEFT improves stability relative to SFT, but still does not match the calibration quality of zero-shot inference. Notably, accuracy and calibration respond differently to adaptation, with accuracy gains rarely translating into improved confidence.

3.3 Fairness Evaluation :

Fairness results in Table 4 reveal a clear accuracy–equity trade-off. Mitra achieves the lowest disparity metrics but at significantly reduced predictive performance. Zero-shot and meta-learning generally offer the best balance of accuracy and fairness, with OrionMSP and OrionBiX maintaining competitive accuracy and moderate disparity levels. SFT exhibits the largest swings in fairness, especially for TabICL and OrionBiX, whereas TabPFN remains comparatively stable across strategies. PEFT variants show mild fairness improvements in some settings but do not consistently outperform zero-shot or meta-learning.

3.4 Key Findings :

Fine-tuning effectiveness is highly model- and data-dependent. TabPFN benefits the most from fine-tuning particularly SFT on medium and wide datasets, while remaining well-calibrated. OrionMSP excels primarily in zero-shot and meta-learning regimes, especially on large or imbalanced datasets. TabDPT achieves strong efficiency–performance trade-offs under PEFT. In contrast, TabICL consistently degrade under SFT, particularly on small or heterogeneous datasets. Overall, fine-tuning yields limited improvements and is most useful in mid-sized or high-dimensional settings. Zero-shot inference remains preferable in low-data regimes and when calibration or fairness stability is important.

Practical Guidelines :

- PREFER ZERO-SHOT FOR SMALL DATA: Zero-shot inference is strongest on small datasets (<1K samples) where fine-tuning often overfits.
- USE FINE-TUNING SELECTIVELY: Apply it mainly on medium-sized (1K–10K) or wide-feature datasets, where TabPFN and TabDPT show consistent benefits.
- META-LEARNING AS DEFAULT: Meta-learning provides modest but reliable gains and preserves calibration, particularly on imbalanced datasets.
- CAUTION WITH SFT: Full supervised fine-tuning frequently reduces accuracy and calibration and should be reserved for models tolerant to full updates (e.g., TabPFN).
- PEFT FOR EFFICIENCY: PEFT avoids overfitting and recovers most useful improvements, making it especially effective for TabDPT and OrionMSP.

Overall, aligning the adaptation strategy with dataset size, feature dimensionality, and calibration requirements is crucial, as zero-shot inference often remains competitive.

3.5 Limitations :

Our evaluation focuses on binary and multi-class classification; PEFT is not supported for some models. Sensitive attributes for fairness are manually defined and vary across datasets. For consistency, results are reported only on the common subset available to all models within each benchmark suite: 155/181 for TALENT, 27/36 for TabZilla, and 63/72 for OpenML-CC18.

4 Conclusion

This work demonstrates that tabular foundation models achieve strong zero-shot performance but fine-tuning provides selective benefits that depend heavily on model architecture and dataset characteristics. While meta-learning and PEFT offer moderate gains in specific scenarios, full supervised fine-tuning often degrades performance and calibration. Our guidelines can help practitioners navigate these trade-offs, emphasizing the importance of understanding when adaptation helps versus when zero-shot suffices.

References

- [1] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, pages 785–794, 2016.
- [2] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural information processing systems*, 30, 2017.
- [3] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. Catboost: unbiased boosting with categorical features. *Advances in neural information processing systems*, 31, 2018.
- [4] Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6):7499–7519, 2024.
- [5] Shriyank Somvanshi, Subasish Das, Syed Aaqib Javed, Gian Antariksa, and Ahmed Hossain. A survey on deep tabular learning. *arXiv preprint arXiv:2410.12034*, 2024.
- [6] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep learning models for tabular data. *Advances in neural information processing systems*, 34:18932–18943, 2021.

- [7] Yury Gorishniy, Ivan Rubachev, Nikolay Kartashev, Daniil Shlenskii, Akim Kotelnikov, and Artem Babenko. Tabr: Tabular deep learning meets nearest neighbors. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024.
- [8] Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.
- [9] Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeyer, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nat.*, 637(8044):319–326, 2025.
- [10] Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. Tabicl: A tabular foundation model for in-context learning on large data. *CoRR*, abs/2502.05564, 2025.
- [11] Mohamed Bouadi, Pratinav Seth, Aditya Tanna, and Vinay Kumar Sankarapu. Orion-msp: Multi-scale sparse attention for tabular in-context learning, 2025.
- [12] Mohamed Bouadi, Pratinav Seth, Aditya Tanna, and Vinay Kumar Sankarapu. Orionbix: Biaxial attention for tabular in-context learning, 2025.
- [13] Junwei Ma, Valentin Thomas, Rasa Hosseinzadeh, Hamidreza Kamkari, Alex Labach, Jesse C. Cresswell, Keyvan Golestan, Guangwei Yu, Maksims Volkovs, and Anthony L. Caterini. Tabdpt: Scaling tabular foundation models. *CoRR*, abs/2410.18164, 2024.
- [14] Xiyuan Zhang, Danielle C. Maddix, Junming Yin, Nick Erickson, Abdul Fatir Ansari, Boran Han, Shuai Zhang, Leman Akoglu, Christos Faloutsos, Michael W. Mahoney, Cuixiong Hu, Huzefa Rangwala, George Karypis, and Bernie Wang. Mitra: Mixed synthetic priors for enhancing tabular foundation models, 2025.
- [15] Chelsea Finn, Pieter Abbeel, and Sergey Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International Conference on Machine Learning (ICML)*, pages 1126–1135. PMLR, 2017.
- [16] Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. *arXiv preprint arXiv:1801.06146*, 2018.
- [17] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2021.
- [18] Haonan Liu et al. Parameter-efficient fine-tuning: A comprehensive survey. *arXiv preprint arXiv:2403.14608*, 2024. Note: Verify exact authors and details.
- [19] Olga Razuvayevskaya et al. Empirical analysis of parameter-efficient fine-tuning methods. *arXiv preprint*, 2024. Note: Verify exact title, authors, and arXiv number.
- [20] Valentin Thomas, Junwei Ma, Rasa Hosseinzadeh, Keyvan Golestan, Guangwei Yu, Maksims Volkovs, and Anthony Caterini. Retrieval & fine-tuning for in-context tabular models. *arXiv preprint arXiv:2406.05207*, 2024.
- [21] Han-Jia Ye, Si-Yang Liu, Hao-Run Cai, Qi-Le Zhou, and De-Chuan Zhan. A closer look at deep learning methods on tabular datasets. *arXiv preprint arXiv:2407.00956*, 2024.
- [22] Bernd Bischl, Giuseppe Casalicchio, Matthias Feurer, Pieter Gijsbers, Frank Hutter, Michel Lang, Rafael G Mantovani, Jan N van Rijn, and Joaquin Vanschoren. Openml benchmarking suites. *arXiv preprint arXiv:1708.03731*, 2017.
- [23] Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Ganesh Ramakrishnan, Vishak Prasad, Micah Goldblum, and Colin White. When do neural nets outperform boosted trees on tabular data? In *Advances in Neural Information Processing Systems*, 2023.
- [24] Aditya Tanna, Pratinav Seth, Mohamed Bouadi, Utsav Avaiya, and Vinay Kumar Sankarapu. Tabtune: A unified library for inference and fine-tuning tabular foundation models. *arXiv preprint arXiv:2511.02802*, 2025.
- [25] Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola. Autogluon-tabular: Robust and accurate automl for structured data. *arXiv preprint arXiv:2003.06505*, 2020.
- [26] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. PMLR, 2017.
- [27] Solon Barocas, Moritz Hardt, and Arvind Narayanan. Fairness and machine learning. *fairmlbook.org*, 2019.
- [28] Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. *Advances in neural information processing systems*, 29, 2016.
- [29] Geoff Pleiss, Manish Raghavan, Felix Wu, Jon Kleinberg, and Kilian Q Weinberger. On fairness and calibration. *Advances in neural information processing systems*, 30, 2017.